

PrepME AI Project Roadmap & Progress Tracker

Project Goal

The goal of this project is not just to build a Notes App.

The actual goal is:

- Understand how real applications are built
- Learn backend development fundamentals
- Learn frontend-backend communication
- Understand authentication and authorization
- Learn database design and CRUD operations
- Gain knowledge expected from a junior-to-mid level developer

By the end of this project, I should be able to confidently explain:

- How login systems work
- How JWT works
- How APIs are built
- How databases are designed
- How frontend communicates with backend
- How real applications are structured

Current Progress

Project Completion:

~70% Complete

Phase 1: Backend Fundamentals COMPLETED

Topics Covered:

Node.js

- Runtime environment for JavaScript
- Runs backend code

Express.js

- Creates APIs
- Handles routes
- Processes requests and responses

MongoDB

- Database
- Stores application data

Mongoose

- Connects Node.js with MongoDB
- Creates schemas and models

Environment Variables

- .env file
- Store secrets and configuration

Project Structure

- routes
- controllers
- models
- middleware
- config

Status:

COMPLETED 

Phase 2: Authentication System COMPLETED

Topics Covered:

Signup

Create user account

Login

Verify email and password

bcrypt

Hash passwords before storing

JWT

Generate authentication tokens

Middleware

Verify JWT before accessing routes

Protected Routes

Allow only authenticated users

Authorization Header

Send JWT with requests

localStorage

Store JWT in browser

Status:

COMPLETED 

Phase 3: Database Relationships & CRUD

COMPLETED

Topics Covered:

CRUD

Create

Read

Update

Delete

MongoDB Methods

save()

findOne()

find()

findById()

findByIdAndUpdate()

findByIdAndDelete()

Relationships

User → Notes

ObjectId

Used to create references

ref

Connect collections together

Status:

COMPLETED 

Phase 4: Security Concepts COMPLETED

Topics Covered:

Authentication

Question:

Who are you?

Authorization

Question:

Are you allowed to do this?

Ownership Validation

User can only edit/delete own notes

Example:

User A cannot delete User B's note

Status:

COMPLETED 

Phase 5: Frontend Integration IN PROGRESS

Current Progress:

React State

useState()

Axios

Frontend API calls

Profile API Integration

Working

Notes API Integration

Started

Status:

IN PROGRESS 

Next Steps (Phase 5 Completion)

1. Logout Feature

Learn:

```
localStorage.removeItem()
```

Flow:

```
Remove Token  
↓  
User Logged Out
```

2. useEffect()

Learn:

```
useEffect()
```

Purpose:

Run code automatically when page loads.

Example:

```
App Opens  
↓  
Fetch Notes Automatically
```

3. Auto Login Persistence

Current:

```
User manually clicks buttons
```

Goal:

```
App Starts
↓
Check Token
↓
Load User Automatically
```

4. Auto Fetch Notes

Current:

```
Click Get Notes
```

Goal:

```
Login
↓
Notes Load Automatically
```

5. Delete Button In UI

Current:

```
Delete works in Postman
```

Goal:

```
Delete from React UI
```

6. Edit Note In UI

Current:

Update works in Postman

Goal:

Update from React UI

Phase 6: Professional Backend Patterns

Topics To Learn:

Validation

Validate incoming requests

Examples:

Empty Title
Invalid Email
Weak Password

Error Handling

Centralized error management

HTTP Status Codes

Examples:

200 Success
201 Created
400 Bad Request
401 Unauthorized
403 Forbidden
404 Not Found
500 Server Error

Cleaner Project Structure

Understanding how larger projects are organized.

Status:

PENDING ⌚

Phase 7: Deployment

Goal:

Put project online.

Topics:

GitHub

Push code

MongoDB Atlas

Cloud database

Render/Railway

Deploy backend

Environment Variables

Production configuration

Result:

Project accessible online

Status:

PENDING ⌚

Phase 8: Optional Advanced Features

Topics:

Search Notes

Search by title

Pagination

Load data in chunks

Sorting

Newest First Oldest First

User Profiles

Additional user information

Role-Based Access

Admin User Manager

Status:

OPTIONAL 

Skills Already Acquired

Backend:

✓ Node.js

✓ Express

✓ MongoDB

✓ Mongoose

✓ JWT

✓ Middleware

✓ CRUD APIs

✓ Authentication

✓ Authorization

Frontend:

✓ React Basics

✓ useState

✓ Axios

✓ API Integration

✓ Conditional Rendering

✓ map()

Database:

✓ Schemas

✓ Models

✓ Collections

✓ Relationships

✓ ObjectId

Final Goal

By the end of this project I should be able to explain:

1. How login works
2. How JWT works
3. How middleware works

4. How CRUD APIs are built
5. How frontend calls backend APIs
6. How MongoDB stores data
7. How authentication differs from authorization
8. How a real full-stack application is structured
9. How to deploy a project
10. How to discuss the project confidently in interviews

Current Progress:

70% Complete

Remaining:

30% Complete

The remaining 30% focuses on making the application production-like rather than educational.